

EXERCISE: 12.1

$$i) \frac{\partial u}{\partial x} = \frac{\partial u}{\partial y} \quad u(x, y) = X(x)Y(y)$$

$$X'Y = XY'$$

$$X'Y - XY' = 0$$

$$X'Y = XY'$$

$$v(x, y) = \log c_1 e^{kx} c_2 e^{ky}$$

Dividing by XY

$$\frac{X'}{X} = \frac{Y'}{Y}$$

$$\frac{X'}{X} = \frac{Y'}{Y}$$

$$\frac{X'}{X} = \frac{Y'}{Y} = k$$

$$\frac{X'}{X} = k \rightarrow$$

$$\log X = kx + \log c$$

$$\log X = kx \log e + \log c$$

$$\log X = \log e^{kx} + \log c$$

$$\log X = \log c e^{kx}$$

$$[X = c e^{kx}]$$

$$\frac{Y'}{Y} = k$$

$$Y$$

$$\log Y = ky + \log c_2$$

$$\log Y = ky \log e + \log c_2$$

$$\log Y = \log c_2 e^{ky}$$

$$(ii) \frac{\partial u}{\partial x} + 3 \frac{\partial u}{\partial y} = 0$$

$$X'Y + 3XY' = 0$$

$$X'Y = -3XY'$$

$$\frac{X'}{X} = -3 \frac{Y'}{Y}$$

$$\frac{X'}{X} = -3 \frac{Y'}{Y} = k$$

$$\frac{X'}{X} = k \rightarrow$$

$$\log X = kx + \log c_1$$

$$\log X = kx \log e + \log c_1$$

$$\log X = \log c_1 e^{kx}$$

$$X = c_1 e^{kx}$$

$$-3 \frac{Y'}{Y} = k$$

$$\frac{Y'}{Y} = -\frac{k}{3}$$

$$\frac{Y'}{Y} = -\frac{k}{3}$$

$$\log Y = \frac{kx}{-3} \log e + \log C_2$$

$$\log Y = \log e^{\frac{kx}{-3}} + \log C_2$$

$$\log Y = \log C_2 e^{-\frac{kx}{3}}$$

$$Y = C_2 e^{-\frac{kx}{3}}$$

$$u(x, Y) = C_1 e^{kx} C_2 e^{-\frac{kx}{3}}$$

$$= C_1 C_2 e^{kx - \frac{kx}{3}}$$

$$= C_1 C_2 e^{\frac{2kx}{3}}$$

$$iii) u_x + u_y = u$$

$$u_x = XY'$$

$$u_y = XY'$$

$$XY' + XY' = XY$$

$$XY' + XY' = XY$$

$$\frac{X'}{X} + \frac{Y'}{Y} = 1$$

$$\frac{X'}{X} = 1 - \frac{Y'}{Y} = k$$

$$\left(\frac{X'}{X}\right) = k$$

$$\log X = k \log e + \log C_1$$

$$\log X = \log C_1 e^k$$

$$X = C_1 e^k$$

$$1 - \frac{Y'}{Y} = k$$

$$1 - k = \frac{Y'}{Y}$$

$$\left(\frac{Y'}{Y}\right) = (1 - k)$$

~~$$\log Y = (1 - k) \log e + \log C_2$$~~

~~$$\log Y = (1 - k) \log e + \log C_2$$~~

$$\ln Y = (1 - k)Y + C_2$$

$$Y = e^{(1 - k)Y + C_2}$$

$$u = X \cdot Y = C_1 e^k \cdot e^{(1 - k)Y + C_2}$$

$$u = e \text{ (constant)}$$

$$iv) u_x = u_y + u$$

$$u = XY$$

$$u_x = X'Y$$

$$u_y = XY'$$

$$X'Y = XY' + XY$$

$$\frac{X'}{X} = \frac{Y'}{Y} + 1 = k$$

$$\frac{X'}{X} = \frac{Y'}{Y} + 1 = k$$

$$\frac{X'}{X} = \int k$$

$$\ln X = kx + c_1$$

$$X = e^{kx+c_1}$$

$$\frac{Y'}{Y} + 1 = k$$

$$\frac{Y'}{Y} = k - 1$$

$$\frac{Y'}{Y}$$

$$\frac{Y'}{Y} = \int k - 1$$

$$\ln Y = (k-1)y + c_2$$

$$Y = e^{(k-1)y+c_2}$$

$$(x, y) = XY$$

$$= e^{kx+c_1} \cdot e^{(k-1)y+c_2}$$

$$= e^{kx+c_1+(k-1)y+c_2}$$

$$= e^C$$

$$= C$$

$$x \frac{\partial u}{\partial x} = y \frac{\partial u}{\partial y}$$

$$\frac{x X' Y}{XY} = \frac{y X Y'}{XY} = k$$

$$\frac{x X'}{X} = \frac{y Y'}{Y} = k$$

$$\frac{x X'}{X} = k$$

$$\int \frac{X'}{X} = \int \frac{k}{x}$$

$$\ln X = k \ln |x| + \ln |c_1|$$

$$X = \frac{c_1^k}{1}$$

$$\frac{y Y'}{Y} = k$$

$$\int \frac{Y'}{Y} = \int \frac{k}{y}$$

$$\ln Y = k \ln |y| + \ln |c_2|$$

$$Y = c_2 y^k$$

$$u = XY = c_1 x^k c_2 y^k$$

$$= c_1 c_2 x^k y^k$$